

LASER INTERFEROMETER GRAVITATIONAL WAVE OBSERVATORY
- LIGO -
CALIFORNIA INSTITUTE OF TECHNOLOGY
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Technical Note	LIGO-T2300319v1-	2023/08/21
Analysis of Gaussian Beam Interference for Detecting Focal Length		
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1 Introduction

The goal of this project is to recreate the pattern from the interference of two laser beams. We will study this with a Michelson interferometer, where the end of one arm holds a flat mirror and the other holds a concave mirror. Reversely, given an image of this interference pattern, we will determine the focal length of this concave mirror.

Ideally, laser beams can be approximated with Gaussian beams. Gaussian beams are called so because of their Gaussian-shaped intensity profile, and they are relatively easy to study. This is relevant for both the laser used in these experiments and the laser used in LIGO.

2 Procedures

2.1 Beam Profiling

The first step in recreating a laser beam is to measure its profile. Using a beam profiler and its software, we measure both the horizontal and vertical radii of the beam. The radius of the beam is defined as where intensity is $1/e^2$ of its maximum at its axis of propagation. As the beam propagates against space, its waist $w(z)$ follows the equation

$$w(z) = w_0 \sqrt{1 + \left(\frac{z - z_0}{z_R} \right)^2},$$

where w_0 is the smallest possible radius occurring at the beam's waist, z is in the direction of propagation, z_0 is the z -axis' location of the waist, and

$$z_R = \frac{\pi w_0^2 n}{\lambda}.$$

For this entire experiment, we use a beam of $\lambda = 633$ nm, and we assume $n \approx 1$. Using SciPy's curve_fit method (least squares regression), we fit our measurements to the beam waist equation (see Fig. 1, Table 1).

Table 1: Original Beam Output Parameters from CurveFit.ipynb

Parameter	X	Y
w_0 (μm)	360.2	357.2
z_0 (mm)	0.03958	11.40
z_R (mm)	644.1	633.3

For the sake of visualizing the beam, we also developed 3DModel.ipynb (see Fig. 2)

Being able to model the unobstructed beam, we then tested the profiling script with biconvex lenses. Fig. 3 shows that adding a lens forms a new Gaussian beam with a waist near the focal point of the lens.

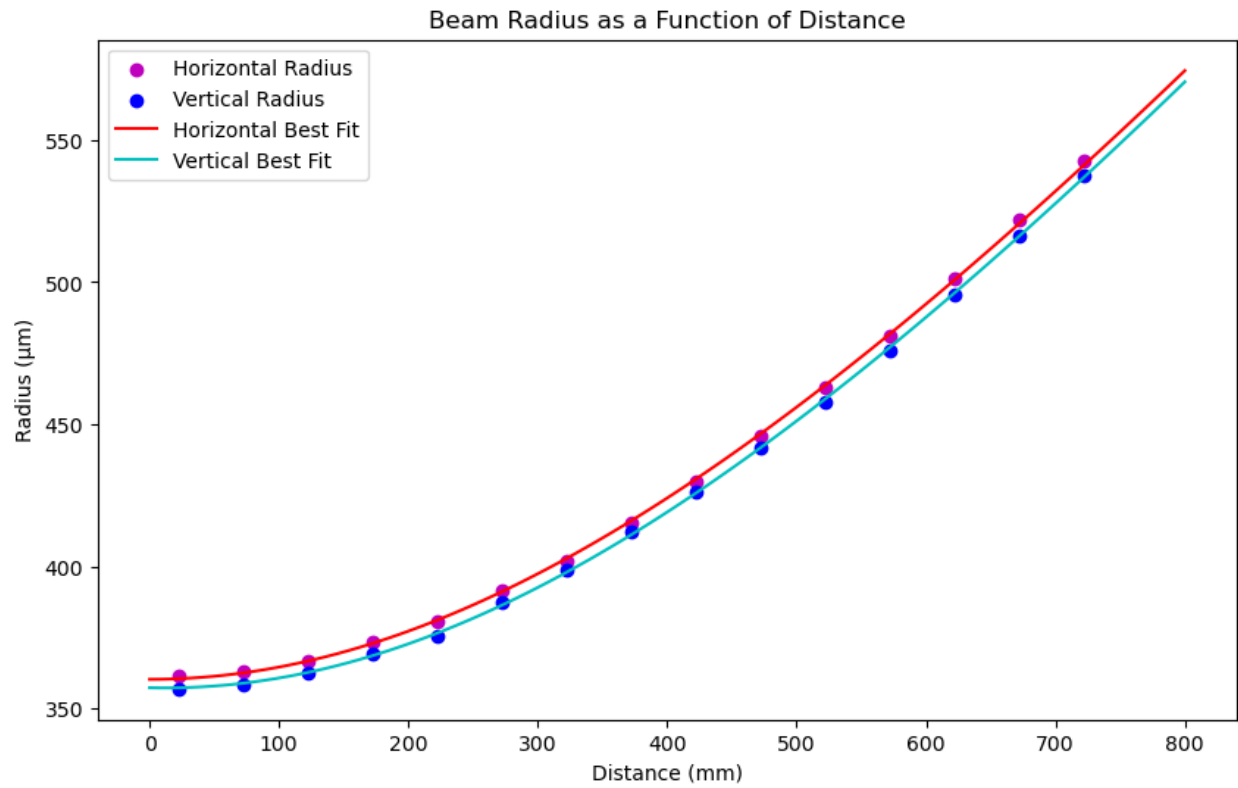


Figure 1: Curve fitting the 633 nm beam, from CurveFit.ipynb

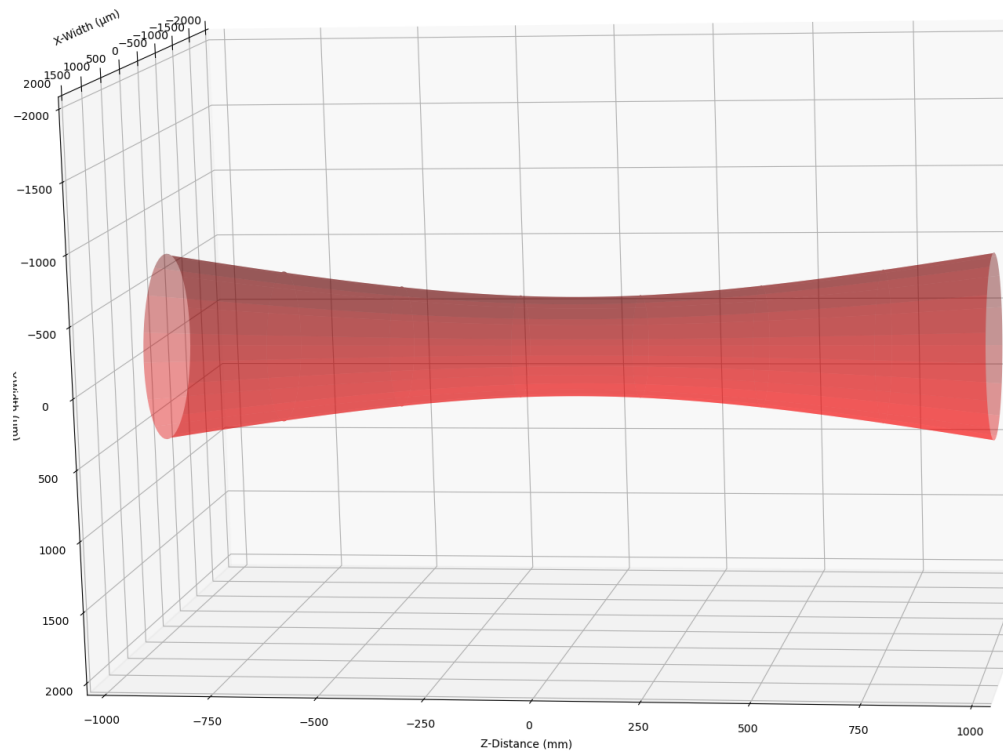


Figure 2: 3D Model of the original beam, from 3DModel.ipynb

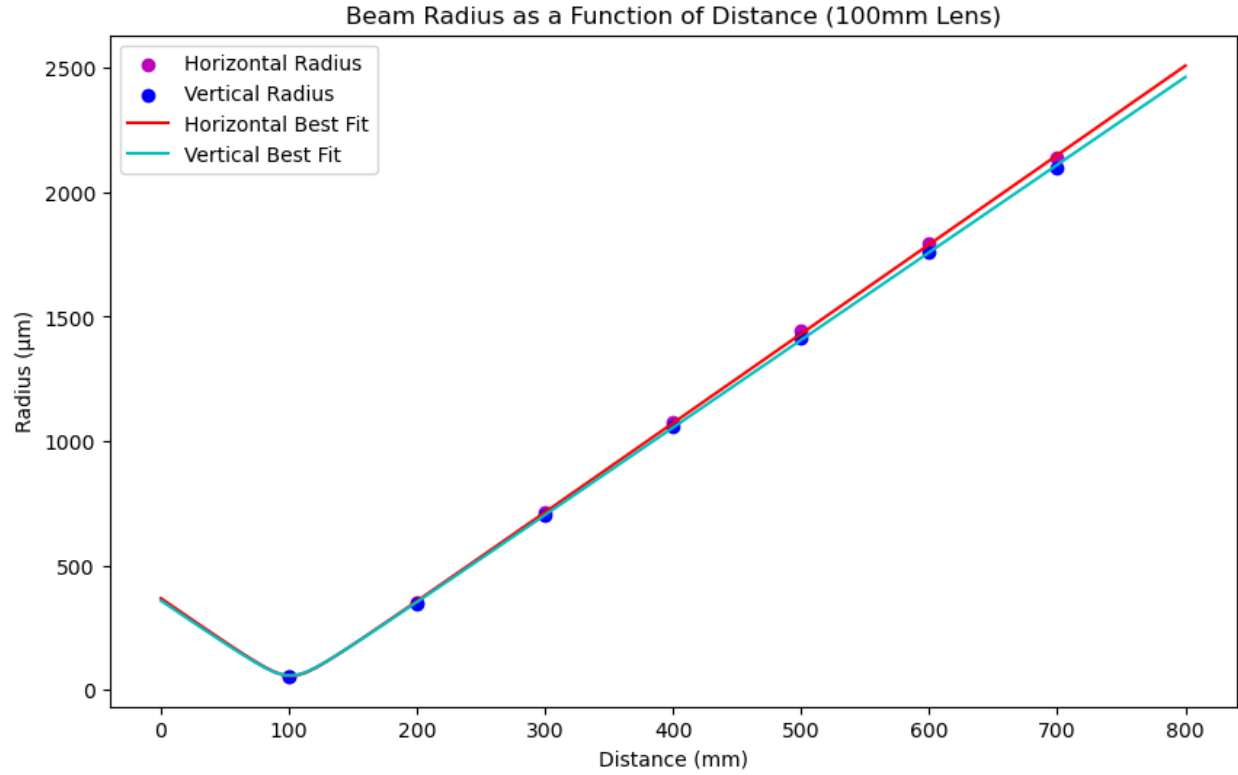
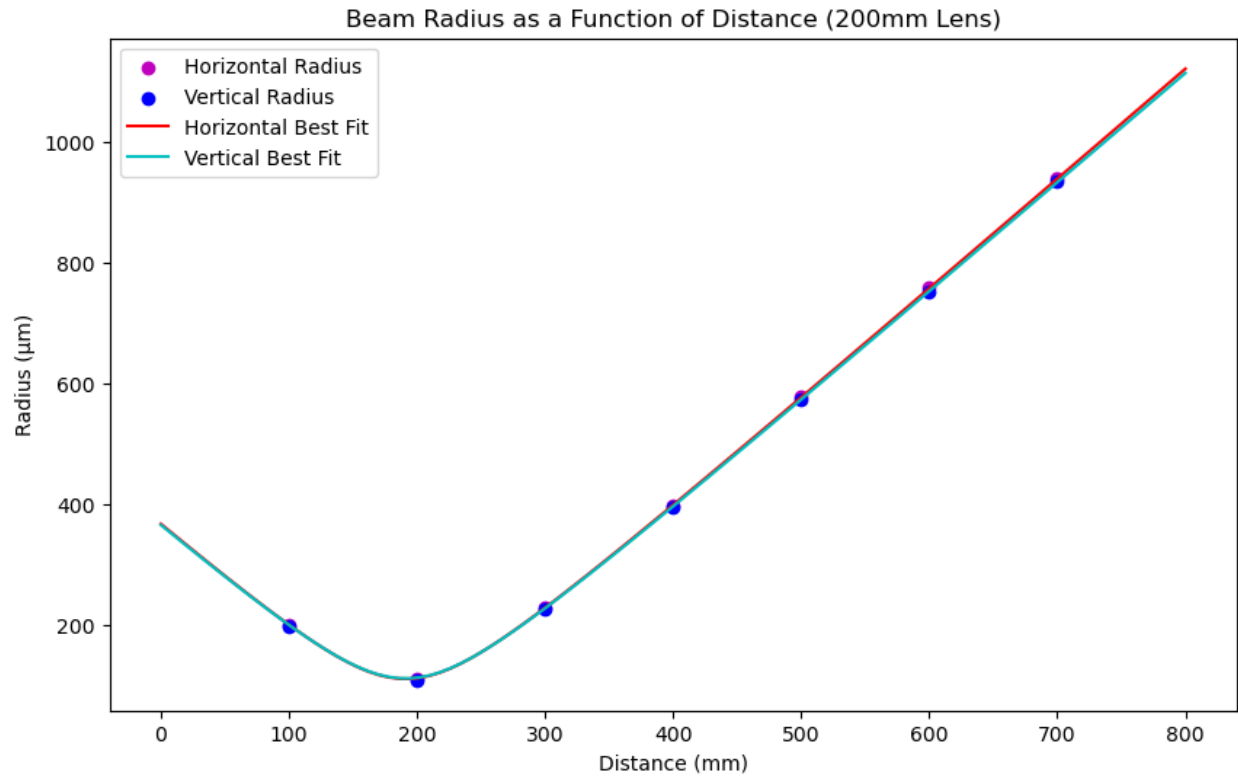
(a) $f = 100.0$ mm(b) $f = 200.0$ mm

Figure 3: Adding lenses to the beam, from CurveFit.ipynb

2.2 Focal Length Detection

Since the interferometer will contain a concave mirror, it will be necessary to study how it will affect the beam. We first examine biconvex lenses, since they can substitute for a concave mirror as long as the focal lengths are equal. Furthermore, biconvex lenses are easier to profile than mirrors.

Already with an understanding of Gaussian beam propagation, all that is further needed to determine the focal length of a lens is the effect of the lens on the beam's propagation. Traditionally, the thin-lens equation is described as follows:

$$\frac{1}{s} + \frac{1}{s'} = \frac{1}{f}.$$

Where s is the distance to the input beam's waist, s' is the distance to the output beam's waist, and f is the focal length. However, Gaussian beams behave uniquely through a lens. For example, if the input beam's waist is one focal length away from the lens, the output beam's waist also is one focal length away from the lens. Additionally, the maximum distance for the output beam's waist occurs when the input beam's waist is at $f + z_R$.

For these reasons, Sidney A. Self's "Focusing of spherical Gaussian Beams" develops an equation to describe a Gaussian beam through a thin lens:

$$\frac{1}{s + \frac{z_R^2}{s-f}} + \frac{1}{s'} = \frac{1}{f}.$$

Utilizing this equation and measurements of the beam both before and after interacting with a 100.0 mm lens, we create FocalLengthDetector.ipynb.

Table 2: Output Parameters from FocalLengthDetector.ipynb

Parameter	w_0 (μm)	z_0 (mm)	z_R (mm)
No lens, before z_{lens} :	358.1	-18.14	636.5
No lens, after z_{lens} :	353.0	-850.9	618.6
With lens :	36.9	81.8	6.759

Where z_{lens} is the location of the lens in the direction of propagation. Additionally, we get an output of Fig. 4 and

Focal length: 98.3 mm

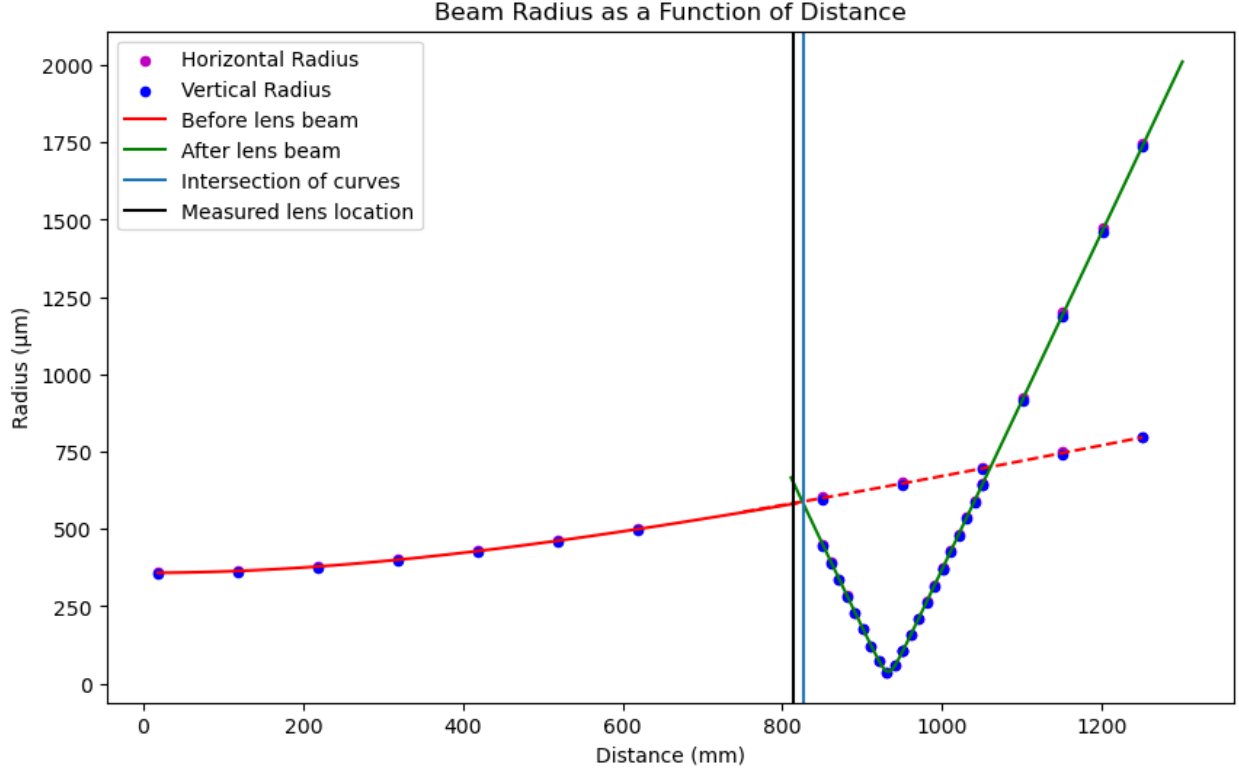


Figure 4: Graphical representation from FocalLengthDetector.ipynb

2.3 Interferometer Setup

Next is to build the interferometer. We are constrained by an optical table of 90 cm by 100 cm, a camera with an image area of 8.98 mm by 6.71 mm, and lenses ranging from 35.0 mm to 750.0 mm in focal length. For this reason, we set goals of the arm length (distance from the beam splitter to the mirrors) equalling 400 mm, a beam radius of <5 mm at the mirrors, and a beam radius of <3 mm at the camera.

We designed Setup.ipynb to model our interferometer (see Fig. 5), using principles of beam propagation and lens interaction. Instead of using the beam waist and lens equations, we opt for using the complex beam parameter $q(z)$ where

$$q(z) = z + iz_R$$

and

$$\frac{1}{q(z)} = \frac{z - iz_R}{z^2 + z_R^2} = \frac{1}{R(z)} - i \frac{\lambda}{n\pi w^2(z)}.$$

Where $R(z)$ is the radius of curvature (the reciprocal of the curvature), but is not used for this project. The reason we use $q(z)$ becomes apparent when performing ray transfer matrix analysis. Propagation through space and through lenses can be described with this simple equation:

$$\begin{bmatrix} q_2 \\ 1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} q_1 \\ 1 \end{bmatrix}$$

Where the ray transfer matrices (ABCD matrices) for propagation through free space and through a thin lens are, respectively

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1/f & 1 \end{bmatrix}.$$

With our program completed, all that was left was to adjust our parameters to fit our constraints. Show in Fig. 5, our parameters were: 20 mm to the 35.0 mm lens, 540 mm to the 400.0 mm lens, 590 mm to the beam splitter, 400 mm from the beam splitter to the mirrors, 400 mm from the beam splitter to the camera, and a curved mirror focal length of 750.0 mm. The output was:

Waist at mirror: 3.984 mm
 Waist at camera: 2.244 mm
 Waist at camera (curved mirror): 2.006 mm

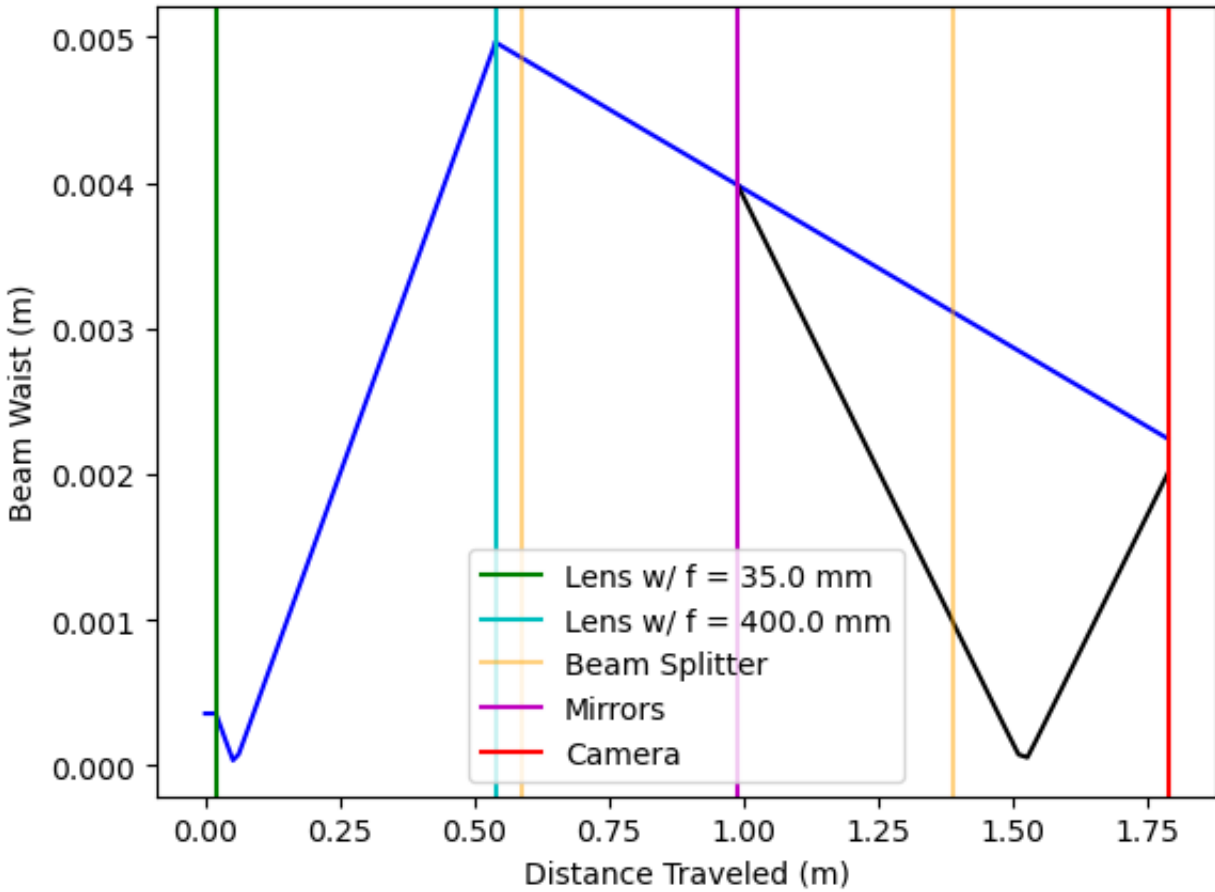


Figure 5: Interferometer setup from Setup.ipynb

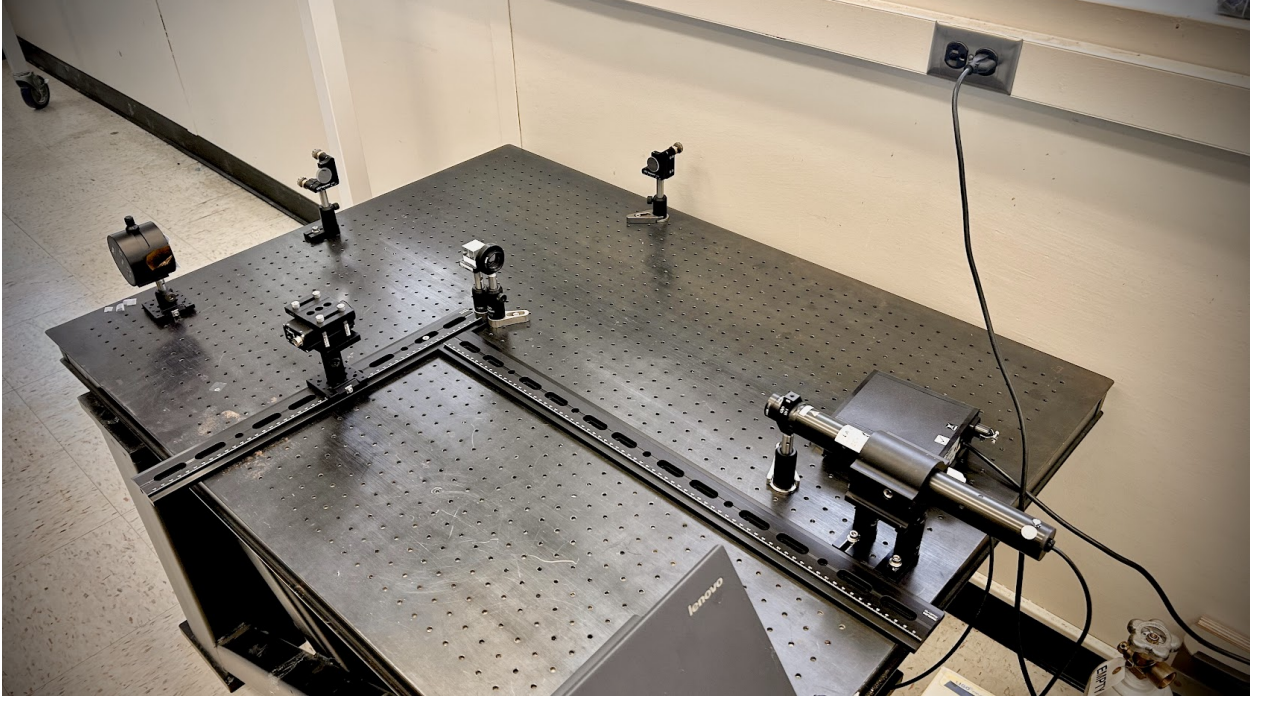


Figure 6: Photo of interferometer

And here is our experimental apparatus. Note that the exact dimensions are not the same as the suggested model, yet principles like the relative location of the lenses remain the same.

2.4 Interference Modeling

We are ready to complete our intended experiment. First, we recreate the fringe pattern from the interference of the curved-mirror beam and the flat-mirror beam. Fig. 7 will be used as the goal image.

We also need to re-profile both beams, since our physical model differs from the suggested one. We measure the beam radii along the propagation direction, and in Interferometer-Beams.ipynb we find the $q(z)$ for 20 mm and 400 mm.

From here, we use these formulas for the electric field (in phasor notation) and the intensity distribution:

$$\mathbf{E}(r, z) = E_0 \frac{w_0}{w(z)} \exp\left(\frac{-r^2}{w(z)^2}\right) \exp\left(-i\left(kz + k\frac{r^2}{2R(z)} - \psi(z)\right)\right)$$

$$I(r, z) = \frac{|\mathbf{E}(r, z)|^2}{2\eta}$$

where $k = 2\pi/\lambda$ is the wave number and $\psi(z) = \arctan\left(\frac{z}{z_R}\right)$ is a phase term called the Gouy phase. Fig. 8 shows the intensity of each beam calculated from these formulas.

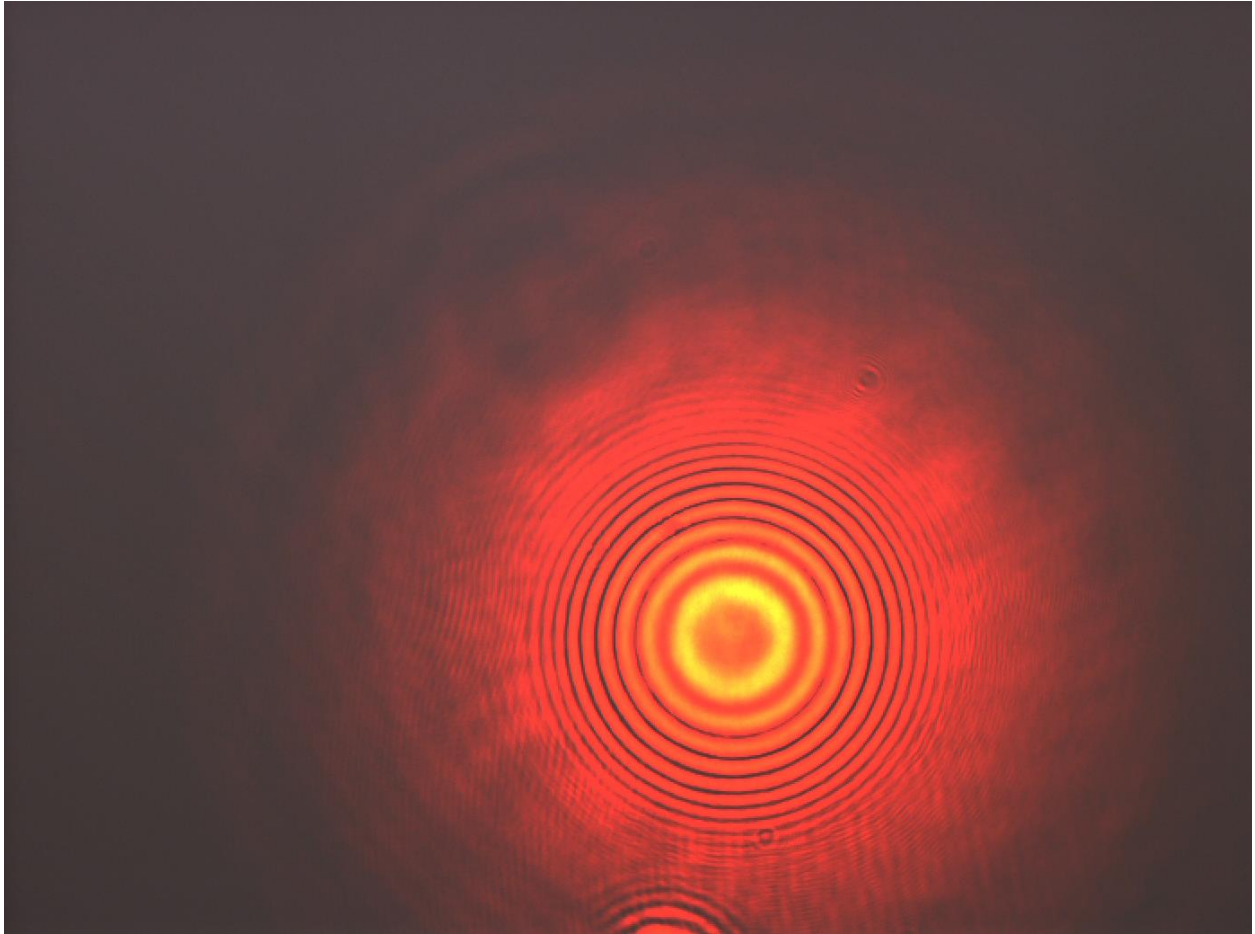


Figure 7: Photo of interference 20 mm away from the beam splitter

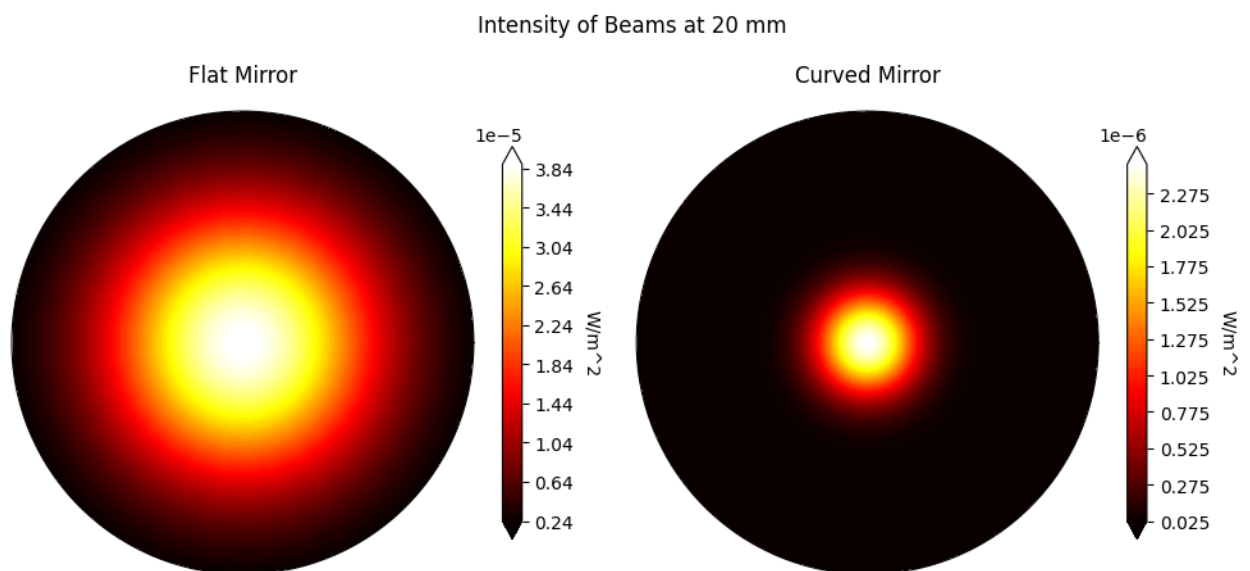


Figure 8: Model of each beam at 20 mm from Simulation.ipynb

To model the interference pattern, we simply add both electric fields then take the intensity. Fig. 9 shows a graph of the interference and Fig. 10 shows the interference models compared with their respective photos, distances of 20 mm and 400 mm away from the beam splitter.

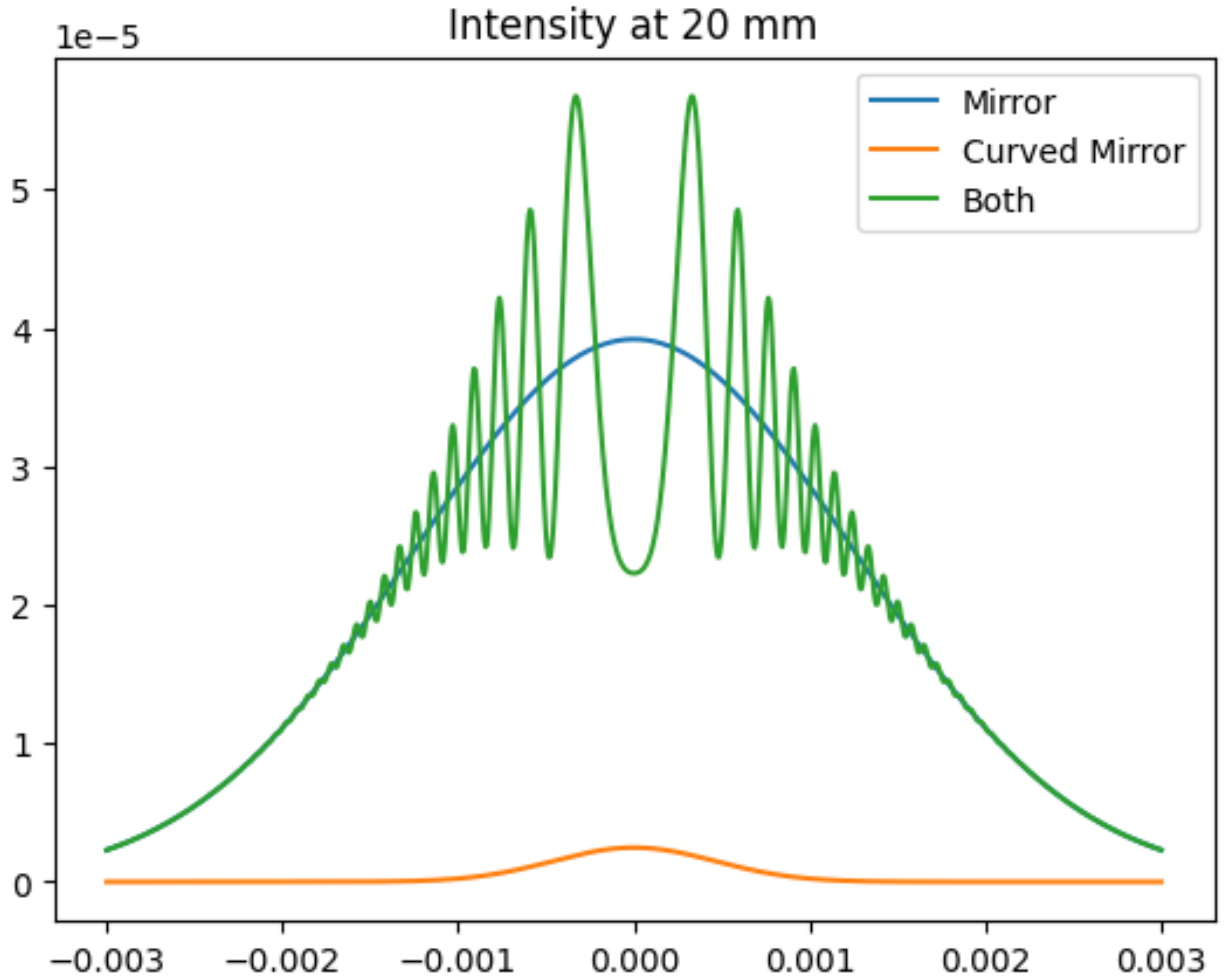
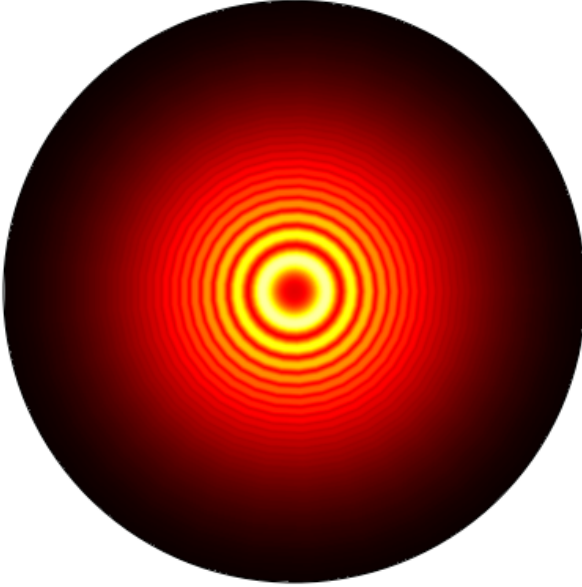
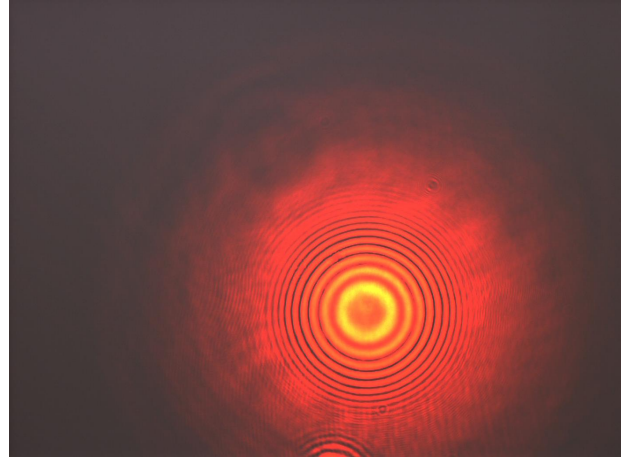


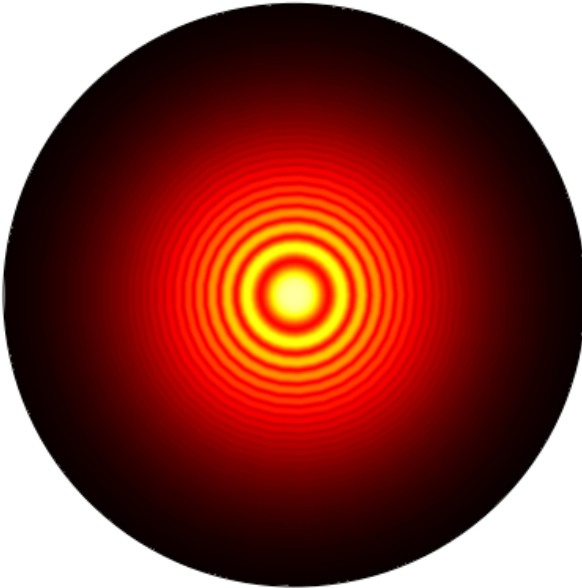
Figure 9: Graph of interference from Simulation.ipynb



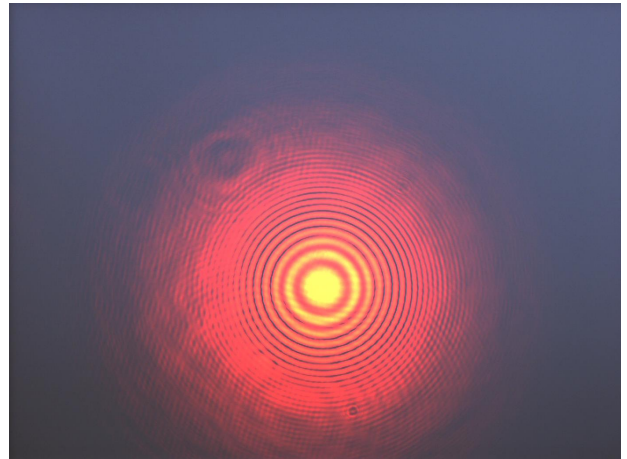
(a) 20 mm model



(b) 20 mm photo



(c) 400 mm model



(d) 400 mm photo

Figure 10: Comparison of the models from Simulation.ipynb with photos

2.5 Interference Image Analysis

Our final project is to work backwards from the photo of the beam interference, determining the focal length of the lens. The methods used to analyze the photo and determine the focal length are located in ImageAnalysis.ipynb. We also used the image 400mm3.jpg (in additional files), since it had the clearest picture of the fringes.

We first find the center of the beam by taking the centroid of the fringes, depicted in Fig. 11. Fig. 12 plots the average pixel intensity based on distance from the center. With the equation we used to model the interference, we fit a curve to the pixel intensity vs. distance graph using the complex beam parameter of the flat beam (from the beam profiler data).

Our results were the complex beam parameter of the curved beam and relative amplitudes of the intensity and electric fields. To achieve a more accurate result, we only used data from within 250 pixels from the center. The complex beam parameter of the curved beam that we obtained from the waist measurements was $0.20 + 0.0079i$. From the photo it was $0.20 + 0.0091i$.

Fig. 13 compares the intensity curve using the fit complex beam parameter of the curved beam (the fitted curve) with the measured intensity data from the photo (the curve it was fit to). Additionally, Fig. 14 compares the fitted curve with the intensity derived from the beam profiler data. The discrepancy in the first figure suggests that the precision of the camera may have limited the accuracy of the measured intensity.

From here, we solve the modified thin-lens equation for a Gaussian beam for the focal length. As parameters, we use the measured distance to the mirrors (400 mm to the beam splitter + 400 mm to the mirrors). Using the photo data, we obtain a focal length of 668 mm. To verify this answer, using the beam profiler data, we obtain a focal length of 670 mm. Interestingly, the listed focal length of the curved mirror is 750 mm.

One thing to note is that the calculated position where the curved beam and flat beam have equal waists is not where the mirror is actually located. The data from the beam profiler predicts a location of the mirrors 70 mm away from their measured distance. The data from the photo predicts a location of the mirrors 20 mm away from their measured distance. Perhaps this can be explained by lurking variables such as physical imperfections in the mirrors and beam splitter, as well as the deviation of the beam from its Gaussian shape.

3 Conclusion

We have demonstrated the ability to model the interference pattern of two Gaussian beams, as well as determining the focal length of the beam affected by a curved mirror. All of the calculations, methods, and functions will be included in the corresponding documentation.

Detected Gaussian Beam Interference Fringes

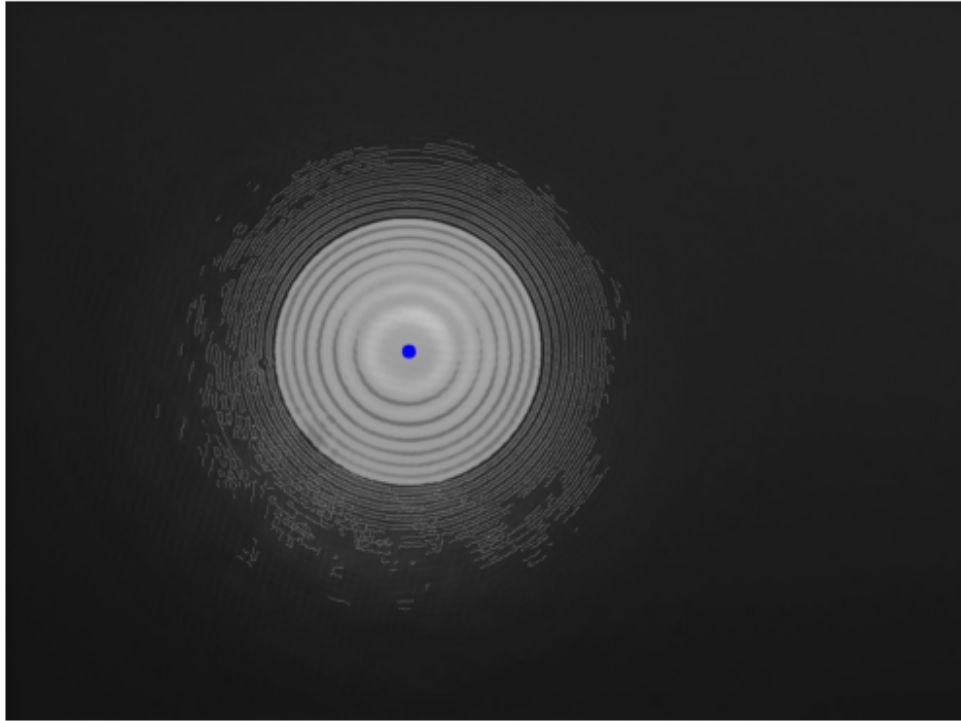


Figure 11: Image's detected fringes and center from ImageAnalysis.ipynb

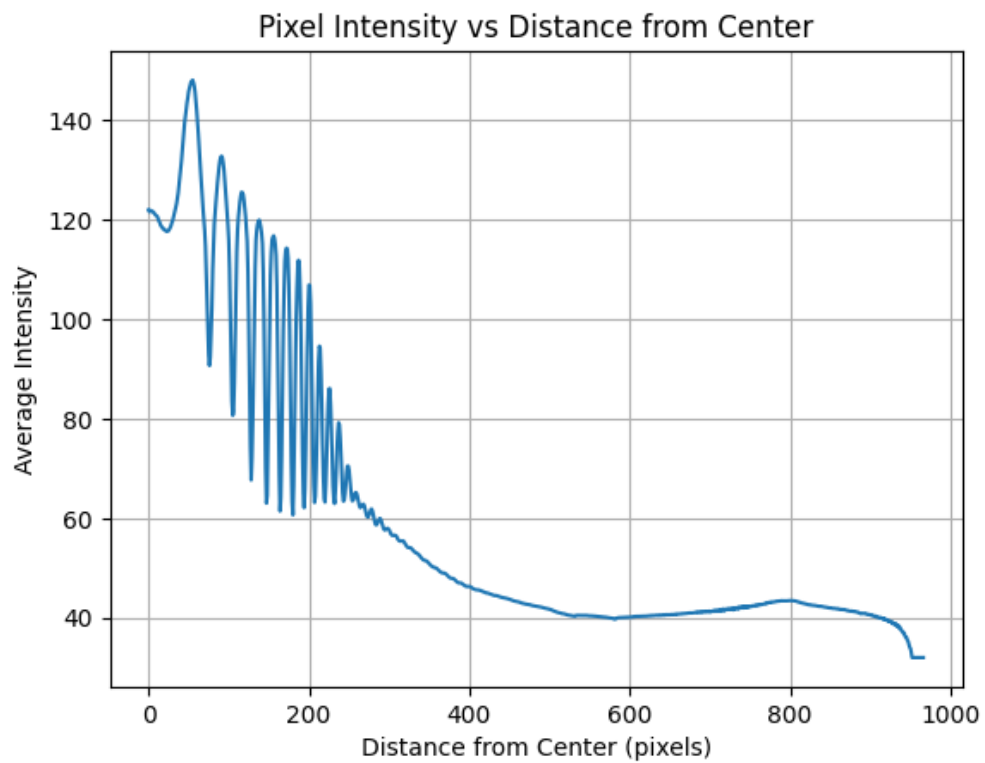


Figure 12: Graph of pixel intensity from ImageAnalysis.ipynb

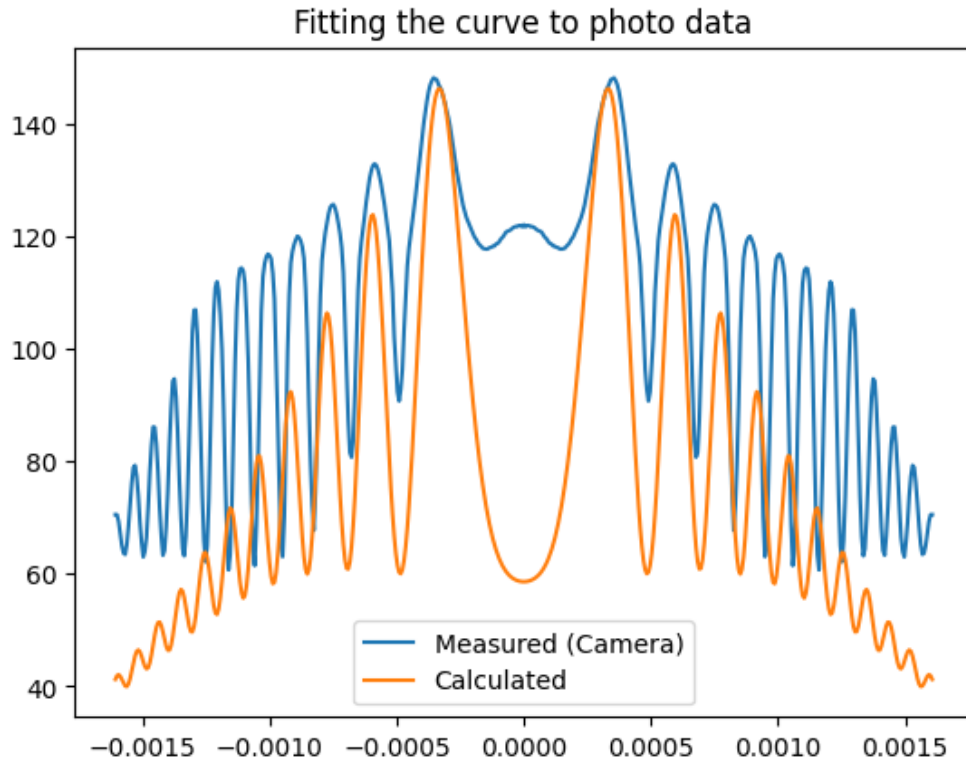


Figure 13: Beam intensity fit to the photo's intensity data from ImageAnalysis.ipynb

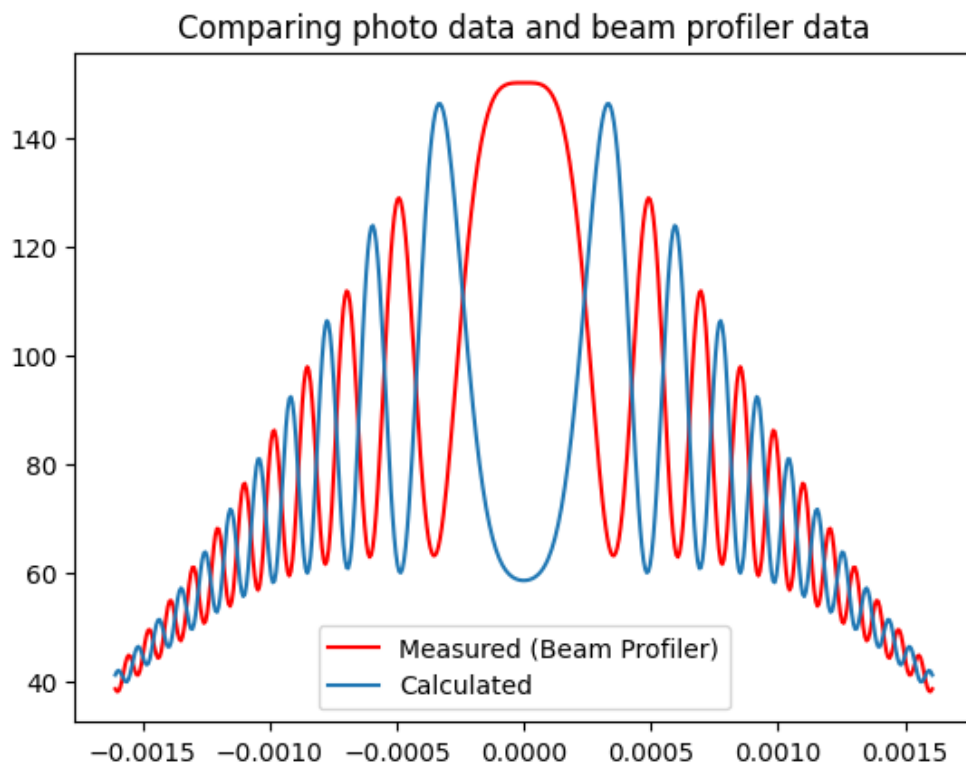


Figure 14: Beam intensity from photo vs. beam profiler from ImageAnalysis.ipynb